

Spirit in Physics:

Word-association response profiles in a five-participant feasibility study

Jun Kawasaki¹, Kazuki Tainaka², Tomonori Takeuchi³

¹Graduate School of Medical and Dental Sciences, Niigata University

²Brain Research Institute, Niigata University, Japan

³Department of Biomedicine, Aarhus University, Denmark

root@junkawasaki.com

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Abstract

We assessed whether repeated word-association latency and skin-potential recordings could be summarized as auditable participant-specific response profiles. Eleven participant directories were present; five provided 970 valid clock-aligned trials over two sessions and 99 shared stimulus words. Pooled test-retest latency correlation across 471 matched participant-word pairs was weak ($r = 0.24$). Mean cross-participant latency correlation was small ($\bar{r} = 0.065$; exploratory two-sided permutation $p = 0.038$), while the first principal-component variance share (27%) was close to its shuffle-null mean (approximately 26%; $p = 0.21$). An in-sample rank-[3, 6, 3] tensor approximation described 75.9% of variance. These observations establish that the available records can be processed into descriptive profiles; they do not validate a psychological construct. The analyzed records contained no matched self-reference and other-target conditions or independent target-specific self-inclusion rating. Therefore this pilot does not measure self-other boundaries. Neither the dimensionless response index nor the shared latency pattern establishes physical energy, a Jungian complex, or a collective unconscious.

1 Introduction

Word-association latency and autonomic activity can vary with the stimulus and the participant [1]. Graph and tensor methods can summarize these multivariate measurements [2], but a constructed geometry has no psychological meaning without reliability and independent validation.

A longer-term motivation of *Spirit in Physics* is to ask whether response patterns could contribute to the study of self-related experience. The immediate empirical question is narrower: can the available word-association and skin-potential records be aligned and summarized reproducibly, and how stable are the measurements? This feasibility report concerns word-linked response profiles, not a test of target-specific self-inclusion. The proposal for matched self/target measurements is documented separately and has not been conducted.

2 Methods

2.1 Ethics and consent

Project materials report approval by Niigata University (approval 2024-0269; March 1, 2025). The study interface required acknowledgment of the research purpose, voluntary participation, the right to withdraw without disadvantage, and audio/video recording before registration.

2.2 Participants and analytic set

Eleven participant directories were present. Six could not enter the complete-record analysis: three continuous-recording files were unavailable git-annex pointers, two directories contained no physiological CSV, and one recording did not overlap the task-event clock. Five participants contributed analyzable synchronized records. This was an availability-based analytic subset, not a random or prospectively selected sample. No demographic or clinical inference is attempted.

2.3 Task and signal extraction

Participants completed two presentations of an approximately 100-word stimulus list. The present analysis uses stimulus identity, event timing, and continuous skin-potential signals; semantic content of spoken association responses is not used in the manifold.

Continuous Mod-002 CSV recordings were sampled at 1 Hz. Response latency was the interval from `response_window_opened` to the first `speech_detected` event; values had to satisfy $0 < \text{latency} < 12$ s. For each channel, skin-potential change ΔSP was the maximum absolute deviation during the response window from the mean of the 3 s preceding stimulus display. A trial was retained only when event times fell inside the recording and all four channel windows were available. This yielded 970 valid trials (198, 181, 195, 199, and 197 by participant).

Channel 2 is the primary skin-potential channel in the current analysis code; Channel 1 is retained as an additional tensor feature. This choice was not preregistered, and electrode-placement metadata sufficient to give the channels a stable physiological interpretation were unavailable.

2.4 Participant-specific response spaces

Within each participant, trials were averaged by stimulus word. The three features were latency, ΔSP_{Ch1} , and ΔSP_{Ch2} . For visualization we define a descriptive response-cost score

$$C(w) = z\{\text{latency}(w)\} + z\{\Delta SP_{Ch2}(w)\}. \quad (1)$$

This dimensionless index combines latency and skin-potential change for display. It is not a measure of self-inclusion, physiological energy expenditure, or information-thermodynamic work.

After feature standardization, squared Euclidean distances generated the radial-basis affinity

$$W_{ij} = \exp\{-\|x_i - x_j\|^2/\sigma^2\}, \quad (2)$$

where σ was the median nonzero pairwise distance. Coordinates were the nontrivial low-order eigenvectors of the symmetric normalized graph Laplacian $L_{sym} = I - D^{-1/2}WD^{-1/2}$. This is a graph-spectral response embedding, not Fisher information geometry.

2.5 Cohort tensor and exploratory statistics

Participant-by-word means were assembled as $\mathcal{T} \in \mathbb{R}^{P \times W \times F}$ with $P = 5$, $W = 99$, and $F = 3$. Missing cells were imputed first by the word-feature mean and then by the feature mean. Each feature was standardized across the participant-word plane. A truncated HOSVD/Tucker approximation [2] used ranks [3, 6, 3]. The graph spectrum, participation ratio, and reconstructed variance are descriptive summaries whose ranks were not selected on an independent sample.

Test-retest stability was Pearson correlation of session-1 and session-2 latency over available matched participant-word pairs. Cross-participant structure was evaluated from the participant-by-word latency matrix. The mean off-diagonal Pearson correlation and first-principal-component variance share were compared with null distributions from 2,000 within-participant word permutations using seed 7. The correlation test was two-sided. These

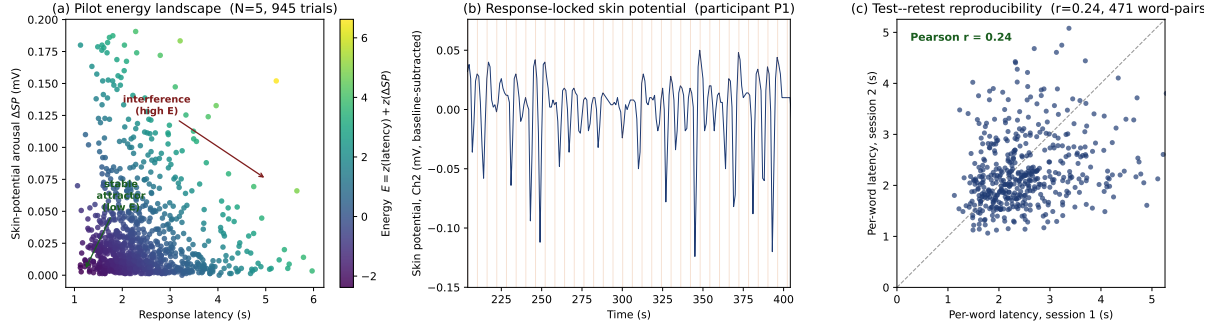


Figure 1: **Pilot measurements.** (a) Response latency and Channel-2 skin-potential change for 970 valid trials, colored by the descriptive response-cost score C . (b) Example continuous skin-potential trace and task events. (c) Session-1 versus session-2 latency for matched participant–word observations.

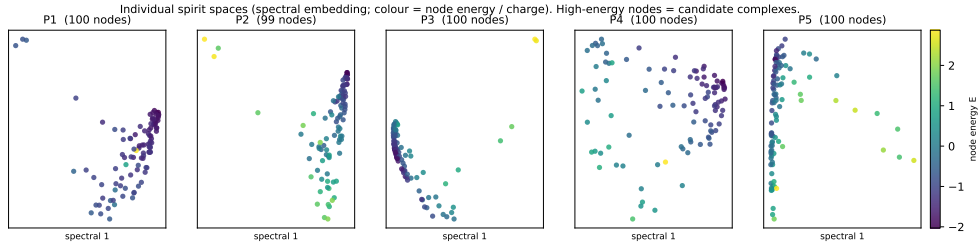


Figure 2: **Participant-specific response spaces.** Each panel is a graph-spectral embedding of one participant’s word nodes from latency and two skin-potential features; color is C . Axes have arbitrary sign and orientation and are not directly comparable without alignment.

analyses were exploratory, were not preregistered, and no multiplicity-adjusted confirmatory claim is made.

3 Results

3.1 Trial-level response landscape

All 970 valid trials are shown in Fig. 1; no percentile trimming is used in the revised display. Because color is computed from the plotted axes, separation of low and high C is expected by construction. It demonstrates measurement coverage, not latent classes. Session-specific latency stability was weak ($r = 0.24$, 471 matched participant–word observations).

3.2 Participant-specific embeddings

The same algorithm produced a two-dimensional response embedding for each participant (Fig. 2). Visual differences are descriptive. No independently assessed personal-salience or complex criterion, or validated classifier, is available in this pilot.

3.3 Cross-participant latency and tensor summaries

Mean off-diagonal cross-participant latency correlation was $\bar{r} = 0.065$ and exceeded the absolute shuffled statistic in the two-sided permutation test ($p = 0.038$). The first principal component explained 27%, close to the approximately 26% shuffle-null mean ($p = 0.21$; Fig. 3). This small exploratory association can reflect common lexical or task effects. The principal-component result does not support a large common component; no Jungian construct was tested.

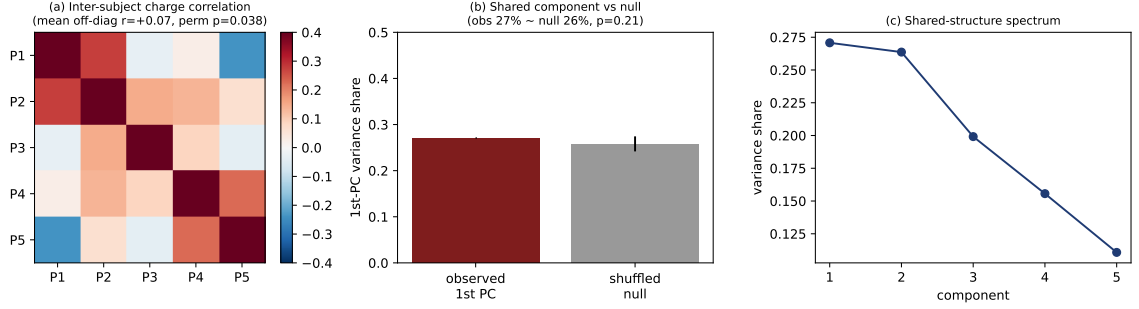


Figure 3: **Exploratory cross-participant latency analysis.** (a) Pairwise correlations and their mean, with a two-sided within-participant permutation test. (b) Observed first-PC variance share versus the shuffle null. (c) Latency variance spectrum. These panels analyze latency, not a physiological “charge.”

The tensor had shape $5 \times 99 \times 3$. The median-distance kernel scale was $\sigma = 4.2109$, the graph eigengap was 0.1948, and the spectrum’s participation ratio was 10.89. Rank-[3, 6, 3] HOSVD reconstructed 75.9% of variance. With only five participants and three features, these values partly reflect dimensional constraints and do not establish a population manifold or a unique “collective coordinate.”

4 Discussion

The analysis shows that the available event and signal records can be transformed into word-linked response embeddings and a cohort tensor. It does not show that distances in these spaces encode closeness to self or another target. The analyzed records did not include repeated matched-word responses under self and other-target instructions or an independent target-specific self-inclusion rating. A future study needs both before interpreting any distance as a candidate self-related measure.

The pooled latency repeatability of $r = 0.24$ is weak and is not an estimate of map or target-distance reliability. The small cross-participant correlation is exploratory, arises from five complete cases after data inspection, and does not establish a collective-unconscious component. The first-PC comparison was null. Latency and skin-potential change, including their standardized sum C , have no calibrated conversion to thermodynamic energy; Landauer’s bound on bit erasure cannot provide one for these observations. The graph is a descriptive embedding, not a Fisher-metric manifold.

A next experiment can compare self and named targets with identical words, collect independent self-inclusion and closeness ratings, and retest on a different day. Distances should be defined from prespecified response profiles, compared with simpler baselines, and assessed independently of the display map. Such an experiment would be a separate study, not a reinterpretation of these five participants.

5 Limitations

The analytic sample was five of eleven participants and depended on local data availability. Missing raw annex objects and absent or misaligned recordings may introduce selection bias. Sampling at 1 Hz limits temporal interpretation. Channel choice was not preregistered, sensor metadata were incomplete, and no negative-control stimulus set or independent validation cohort was available. Semantic association responses and rubber-hand-illusion data were not analyzed. Statistics were exploratory and no adjustment for multiple analytic choices was ap-

plied. No matched self/target condition or independent self-inclusion rating was available in this analysis. Map repeatability and criterion validity were not measured. A separate proposed study has not been preregistered or approved for collection. The current repository is not publicly accessible and the deidentified derived dataset is not yet deposited in a durable public archive.

6 Data and code availability

The analysis code, event-processing logic, and manuscript sources are maintained in the project repository. At the time of manuscript preparation the repository is private and participant-level recordings are not publicly released. A deidentified derived table, versioned code snapshot, environment specification, and checksums have not yet been deposited in a durable archive; the present analysis should therefore not be described as independently reproducible. Access to underlying research data is subject to ethics approval, consent, and an appropriate data-use process; inquiries may be directed to the corresponding author.

7 Earlier public version

An earlier, non-peer-reviewed project website article and PDF presented a broader thermodynamic interpretation and stronger claims about identified structures. This manuscript substantially revises those claims: no physical energy quantity, Jungian complex, or collective-unconscious structure was validated by the pilot. It is related public material and is superseded by the present analysis. See <https://spirit-in-physics.org/paper>.

8 Conclusion

This five-participant pilot demonstrates the feasibility of producing descriptive word-association response profiles from the available records. Weak latency repeatability, a null principal-component comparison, missing sensor provenance and the absence of self/target conditions prevent a validated claim about self-inclusion, physical energy or collective psychology. The next empirical step is a separately approved, matched self/target study with independent ratings and different-day retest.

References

- [1] Carl Gustav Jung. The association method. *The American Journal of Psychology*, 21(2):219–269, 1910.
- [2] Tamara G. Kolda and Brett W. Bader. Tensor decompositions and applications. *SIAM Review*, 51(3):455–500, 2009.